

K/U	A	T	C	Total
60	18	16	6	100

Unit 5 Test

Modified True/False: [5]

Indicate whether the sentence or statement is true or false. If false, change the identified word or phrase to make the sentence or statement true.

- _____ 1. Oxidation processes always result in a *decrease* in the oxidation number of a species.

- _____ 2. In a redox table, the strongest reducing agent is at the *bottom left side* of the table.

- _____ 3. The strongest oxidizing agent present in an electric cell always undergoes a *reduction* at the cathode.

- _____ 4. The cathode of an electrolytic cell is the *positive* electrode. _____
- _____ 5. The electrolysis of water using inert electrodes produces *hydrogen gas at the anode and oxygen gas at the cathode*. _____

Multiple Choice [10]

Identify the letter of the choice that best completes the statement or answers the question.

- _____ 6. In the reaction

$$2\text{HNO}_{3(aq)} + \text{Cu}_{(s)} + 2\text{H}^{+}_{(aq)} \rightarrow 2\text{NO}_{2(g)} + \text{Cu}^{2+}_{(aq)} + 2\text{H}_2\text{O}_{(l)}$$

Which of the following statements is correct?

a.	HNO_3 is the reducing agent, and Cu is the oxidizing agent
b.	HNO_3 is the reducing agent, and H^+ is the oxidizing agent
c.	Cu is the reducing agent, and H^+ is the oxidizing agent
d.	Cu is the reducing agent, and HNO_3 is the oxidizing agent
e.	H^+ is the reducing agent, and Cu is the oxidizing agent

- _____ 7. The oxidation number of sulfur (S) in Na_2SO_3 is

a.	+6	d.	+2
b.	+4	e.	0
c.	+3		

- _____ 8. Which of the following equations does not represent an oxidation-reduction reaction?

a.	$\text{Mg}_{(s)} + 2\text{HCl}_{(aq)} \rightarrow \text{MgCl}_{2(aq)} + \text{H}_{2(g)}$
b.	$\text{CH}_{4(g)} + 2\text{O}_{2(g)} \rightarrow \text{CO}_{2(g)} + 2\text{H}_2\text{O}_{(g)}$

c.	$SO_{3(g)} + CaO_{(s)} \rightarrow CaSO_{4(s)}$
d.	$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}$
e.	$PCl_{3(g)} + Cl_{2(g)} \rightarrow PCl_{5(g)}$

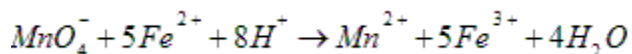
9. Experiments were performed with four strips of metals A , B , C , and D , and their corresponding nitrate solutions $A(NO_3)_3$, BNO_3 , $C(NO_3)_2$, and $D(NO_3)_2$. The following observations were obtained:

	$A_{(s)}$	$B_{(s)}$	$C_{(s)}$	$D_{(s)}$
$A^{3+}_{(aq)}$	no reaction	no reaction	no reaction	no reaction
$B^{+}_{(aq)}$	precipitate	no reaction	precipitate	precipitate
$C^{2+}_{(aq)}$	precipitate	no reaction	no reaction	no reaction
$D^{2+}_{(aq)}$	precipitate	no reaction	precipitate	no reaction

Which of the following is the correct list of the metal ions in order of decreasing strength as oxidizing agents (strongest oxidizing agent first)?

a.	$B^{+} D^{2+} C^{2+} A^{3+}$	d.	$A^{3+} C^{2+} D^{2+} B^{+}$
b.	$A^{3+} D^{2+} C^{2+} B^{+}$	e.	$C^{2+} B^{+} A^{3+} D^{2+}$
c.	$D^{2+} A^{3+} B^{+} C^{2+}$		

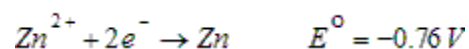
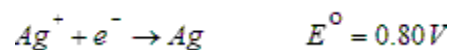
10. Consider the following reaction:



In this reaction, the reducing agent is

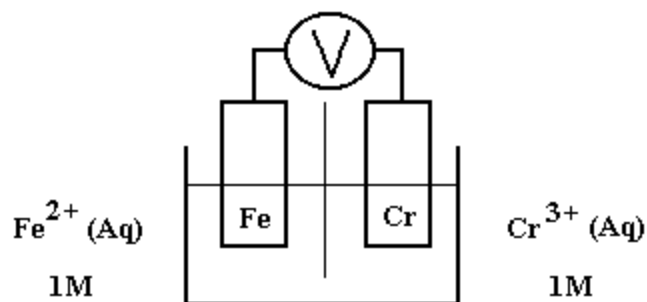
a.	MnO_4^{-}	d.	Mn^{2+}
b.	Fe^{2+}	e.	Fe^{3+}
c.	H^{+}		

11. The following half reactions are involved in an electrochemical cell:

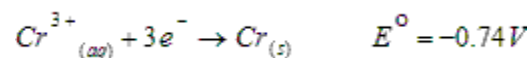
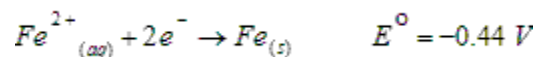


The addition of sodium chloride solution, $NaCl_{(aq)}$, to the silver half-cell results in the precipitation of silver chloride, $AgCl$. Which of the following voltage changes will occur as $NaCl_{(aq)}$ is added to the silver half-cell?

a.	the cell voltage will decrease	d.	the cell voltage will not change
b.	the cell voltage will drop to zero	e.	the cell voltage will double
c.	the cell voltage will increase		



12. Given the following half-cell reactions:



At 25°C, the initial cell voltage of the cell shown in the diagram above is

a. +1.18 V	d. +0.30 V
b. -0.30 V	e. +1.34 V
c. -1.18 V	

13. When inert electrodes are used in the electrolysis of sodium sulfate solution, $Na_2SO_{4(aq)}$, the relevant half-cell reactions are given below

I.	$2H_2O_{(l)} + 2e^{-} \rightarrow H_{2(g)} + 2OH^{-}_{(aq)} (10^{-7} mol/L) \quad E^{\circ} = -0.41 V$
II.	$Na^{+}_{(aq)} + e^{-} \rightarrow Na_{(s)} \quad E^{\circ} = -2.71 V$
III.	$2H_2O_{(l)} \rightarrow O_{2(g)} + 4H^{+}_{(aq)} (10^{-7} mol/L) + 4e^{-} \quad E^{\circ} = -0.815 V$
IV.	$2SO_4^{2-}_{(aq)} \rightarrow S_2O_8^{2-}_{(aq)} + 2e^{-} \quad E^{\circ} = -2.01 V$

Which of these half-cell reactions are **possible** cathode reactions?

a. I and II	d. III and IV
b. I and III	e. I and IV
c. II and IV	

14. Which of the following statements about electrolytic cells is false?

a.	an electrolytic cell can not operate without energy provided by an electrochemical cell
b.	the anode of an electrolytic cell is the negative electrode
c.	the anode of an electrolytic cell is where oxidation occurs
d.	the cathode of the electrolytic cell is where reduction occurs
e.	the electrolytic process is a non-spontaneous reaction

15. Calculate the current applied to an electrolytic cell to transfer 0.50 mol of electrons while the cell is operating for 30.0 min.

a.	27 A	d.	10 A
b.	20 A	e.	5.0 A
c.	15 A		

Short Answers:

- What is oxidized and what is reduced in the following reaction: $\text{N}_2 + 2\text{O}_2 \rightarrow 2\text{NO}_2$ [1]
- Determine the oxidation number of each element in the following substances: [8]
 - H_2SO_4
 - $\text{Ca}_3(\text{PO}_4)_2$
 - Cl_2
 - KMnO_4
 - H_3AsO_4
 - CaSnO_3
- Balance the following equations (use the oxidation number method): [19]
 - $\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$ (acidic solution)
 - $\text{HNO}_3 + \text{H}_2\text{S} \rightarrow \text{NO} + \text{S}$
 - $\text{Cr}_2\text{O}_7^{2-} + \text{Cl}^- \rightarrow \text{Cr}^{3+} + \text{Cl}_2$ (acidic solution)
 - $\text{Cr}(\text{OH})_3 + \text{ClO}_3^- \rightarrow \text{CrO}_4^{2-} + \text{Cl}^-$ (basic solution)
 - $\text{Zn (s)} + \text{SO}_4^{2-} + \text{H}^+ \rightarrow \text{SO}_2 \text{ (g)} + \text{Zn}^{2+} + \text{H}_2\text{O}$
- Iodine normally has oxidation numbers of -1 and 0. Could I_2 serve as a reducing agent? Explain. [2C]
- Describe what happens when chlorine in water is mixed with bromide ions and mineral oil (a non-polar solvent). Explain why this happens. [2C]
- Should chlorine gas be passed through copper tubing? Explain your answer using E° values. [2C]
- Balance the following equations using the half-reaction method. [9]
 - $\text{Ba (s)} + \text{Cl}_2 \rightarrow \text{Ba}^{2+} + \text{Cl}^-$
 - $\text{MnO}_4^- + \text{NO} + \text{OH}^- \rightarrow \text{Mn}^{2+} + \text{NO}_2^- + \text{H}_2\text{O}$
 - $\text{I}^- + \text{NO}_3^- + \text{H}^+ \rightarrow \text{I}_2 + \text{NO}_2 + \text{H}_2\text{O}$
- For each set of reactants, write an equation for the reaction. Use E° values to determine if the reaction is spontaneous. [22]
 - $\text{Mg}_{(\text{s})} + \text{Al}^{3+}$
 - $\text{Hg}^{2+} + \text{Ag}_{(\text{s})}$
 - $\text{Cl}^- + \text{I}_{2(\text{s})}$
 - $\text{Sn}_{(\text{s})} + \text{H}^+$
 - $\text{H}_{2(\text{g})} + \text{Cl}_{2(\text{g})}$
 - $\text{Cu}_{(\text{s})} + \text{I}^-$
- Name two substances which oxidize Ag to Ag^+ . [2]
- A current of 156A is applied to an electrolytic cell containing molten CuCl_2 for 45 min. What substance and how many grams of it will form on the cathode? [5]

11. How many moles of electrons must have been added to an electrolytic cell containing Cd^{2+} ions if 10.5g of Cd(s) forms on the cathode? [5]
12. Draw a diagram of an electrolytic cell for the electrolysis of MgCl_2 . Label all parts of the cell and indicate the reactions occurring at each electrode. Estimate the voltage required to electrolyze MgCl_2 . [8]